Hanyeol Ryu

Research Assistant

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RESEARCH INTERESTS

- Low-power wireless communications
- Channel modeling & estimation
- AI/ML techniques for intelligent wireless communications
- Wireless sensing
- Hardware-Software co-designed IoT applications

EDUCATION

Pusan National University

B.S. in Electronics Engineering Feb 2026

- **PI**: Prof. Sangkil Kim

RESEARCH EXPERIENCE

Pusan National University

Undergraduate Researcher (PI: Dr. Sangkil Kim) Jan 2024 – Feb 2026

- Develop and evaluate energy-efficient backscatter communication systems through hardware–software co-design, enhancing channel reliability and enabling scalable multimedia-oriented IoT applications.

Research Assistant (PI: Dr. Sangkil Kim) Mar 2026 – Present

- Investigate deep learning-driven channel estimation and signal recovery techniques for ultra-low-power backscatter and integrated sensing-communication systems under severe hardware constraints.

PUBLICATIONS

Published Articles (†: First author, *: Corresponding author)

- **J3:** N. Ha, **H. Ryu**, H. Jeong, H. Kim, G. Kim, A. Georgiadis, M. M. Tentzeris, and S. Kim*, "Integrated Sensing and Backscatter Communication (ISABC): Toward Self-Sustainable Autonomous IoT Networks," *IEEE Microwave Magazine,* Accepted, Jun. 2026.
- **J2**: **H. Ryu**†, and S. Kim*, "Ultra-low-power monostatic backscatter platform with phase-aware channel estimation and system-level validation," *IEEE Internet Things J.*, early access, 2026, doi: 10.1109/JIOT.2026.3696951.
- **J1**: J. Choi†, **H. Ryu**†, and S. Kim*, "Improved Backscattering Communication Range Using Reflection Amplifiers," *J. Korean Inst. Electromagn. Eng. Sci,* vol. 36, no. 2, pp. 266-273, Feb. 2025.
- **C2**: **H. Ryu**†, and S. Kim*, "Deep Learning-Aided Phase-Aware Channel Estimation for Error-Floor Suppression over Phase-Drifting Backscatter Links," in *Proc. Int. Symp. Intell. Signal Process. Commun. Syst.*, Accepted, 2026.
- **C1**: **H. Ryu**†, and S. Kim*, "High-Isolation Yagi-Uda Antenna Array using Split-Ring Resonators for Practical Backscatter System," in *Proc. IEEE Antenna Propag. Int. Symp.*, 2025, pp. 2877-2879.

PATENTS

- Korean Patent: *"Residual Phase-Aware Channel Estimation Method for Monostatic Backscatter Communication Systems"*, Apr 2026.

CONFERENCE PRESENTATIONS

Title: Ultra-low-power Backscatter Platform with Phase-Aware Channel Estimation (Oral)

- The Korean Institute of Electromagnetic Engineering and Science (KIEES) Feb 2026

Title: High-Isolation Yagi-Uda Antenna Array using Split-Ring Resonators for Practical Backscatter System (Oral)

- IEEE International Symposium on Antennas and Propagation and USNC-URSI Radio Science Meeting (AP-S/URSI) Jul 2025

Title: Improved Backscattering Communication Range Using Reflection Amplifiers (Poster)

- The Korean Institute of Electromagnetic Engineering and Science (KIEES) Feb 2025

HONORS & AWARDS

- 2nd Place: Hackathon (Theme: Improving Accessibility for People with Disabilities and the Elderly), **Pusan National University**, Aug 2023.
- 2nd Place: Capstone Design Project, **Pusan National University**, Nov 2025.

SERVICE & EXTRACURRICULAR ACTIVITIES

University

Pusan National University

*Treasurer, **Student Council*** Dec 2021 – Dec 2022

*Student Ambassador, **Pusan National University*** Apr 2021 – Feb 2023

*Volunteer, **Overseas Volunteering Program (3D Printer & Arduino) — Thailand*** Dec 2022 – Feb 2023

Military Service

Republic of Korea Marine Corps

*Squad Leader, **ROK Marine Corps Force Reconnaissance*** Feb 2017 – Jul 2018

2nd Place*, **ROK Marine Corps Force Reconnaissance Selection and Qualification Training*** Jun 2017 – Aug 2017

SKILLS & TECHNIQUES

- **Programming:** MATLAB, Python, C++
- **AI/ML:** PyTorch, TensorFlow/Keras
- **RF/EM Simulation:** Ansys HFSS, Advanced Design System (ADS)
- **SDR Platform:** GNU Radio
- **Hardware Design:** OrCAD PCB Editor, OrCAD Capture
- **3D Modeling:** Rhino 7